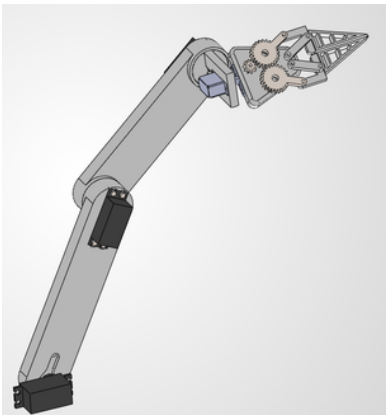
NASHWIN MENDONCA

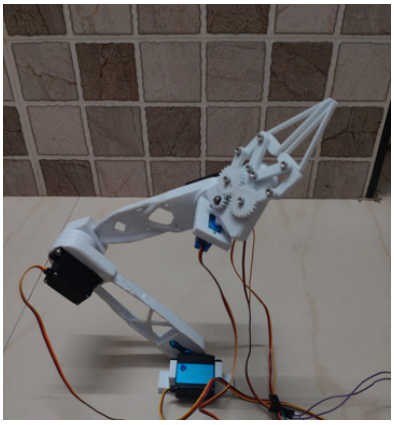
MECHANICAL ENGINEERING STUDENT | NIT SURAT

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ROBOTIC ARM STRUCTURAL OPTIMIZATION







WHAT?

- Designed and fabricated a robotic arm
- High structural stiffness for less material

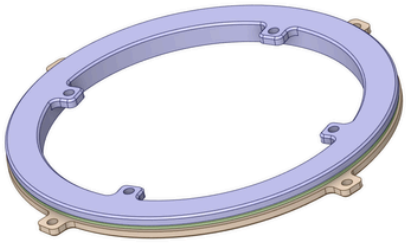
HOW?

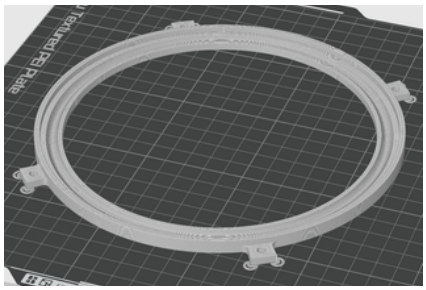
- Used Solidworks for initial cad
- Applied ANSYS (Topology optimization)
- 3D printed optimized parts

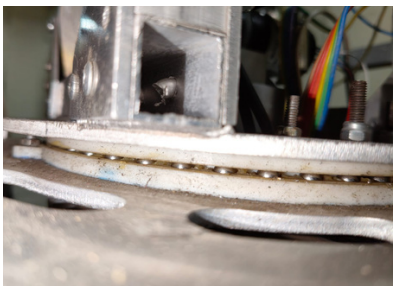
RESULTS

- The arms structural weight dropped by 78%
- The resulting structural stiffness dropped by only 18%

CUSTOM THRUST BEARING







WHAT?

- Designed and abuse tested thrust bearing
- Dimensions 110×110×10 mm
- Target operating time: 150 h

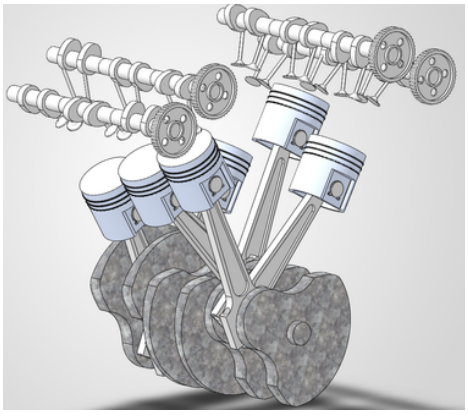
HOW?

- Conducted ANSYS creep and static structural analysis
- 3D printed the bearing guids

RESULTS

- Withstood an axial load of ~196 N.
- There was minute amount of deformation in the raceway under 100 hrs of operation

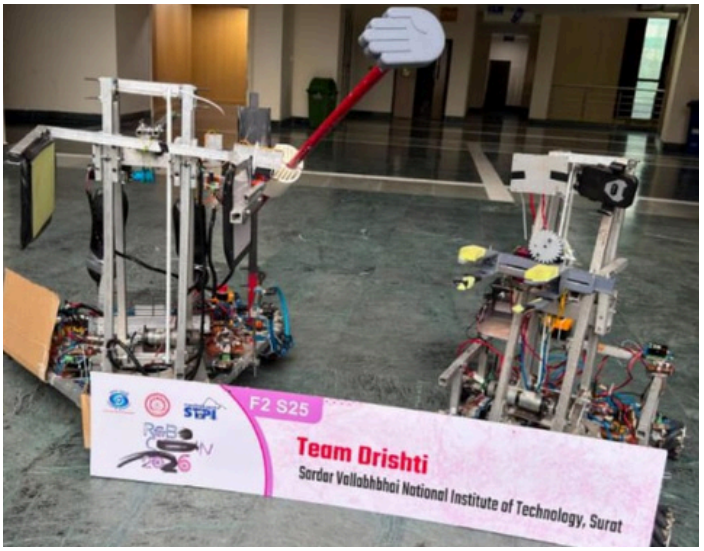
DOHC V6 CAD & Motion Study



- Designed and assembled a DOHC V6 engine (60° bank) in SolidWorks with accurate valve timing, firing order, and combustion-cycle sequencing
- Validated crankshaft, camshaft, piston, and valvetrain synchronization through motion studies

DD ROBOCON -2026

- Designed 2 cooperative robots with 160+ components, 15+ assemblies, and 5+ iterations.
- Owned fabrication of 60+ 3D-printed parts, optimizing orientation, tolerances, and print parameters.
- Applied GD&T to laser-cut components and gussets to ensure dimensional accuracy and assembly fit.
- Redesigned the rubber wheel mechanism into a PETG thrust-bearing system, reducing friction, wobble, and motor heating.
- Applied DFM and creep analysis for reliable 3D-printed mechanical components.
- Collaborated with the ECE team for mechanical-electrical system integration.
- Achieved 4th place nationally among 130 teams at DD Robocon.



Certified Solidworks Professional


DASSAULT SYSTEMES

SOLIDWORKS

CERTIFICATE

Dassault Systèmes confers upon

NASHWIN MENDONCA

the certificate for

SOLIDWORKS Design Professional (CSWP)

June 8 2026

DESIGN

Manish KUMAR
SOLIDWORKS CEO
R&D Vice President

C-SMK8QZ2ULB